
Chapter 3: Coordinate Geometry

A Level Pure Maths Notes · Rexponential

Why this chapter matters

Coordinate geometry translates between algebra and pictures: every equation in this chapter describes a specific shape on the xy -plane, and every question can be attacked either by pure algebraic manipulation or by reasoning from the picture. The strongest students move fluently between both — sketching first to see what's being asked, then using algebra to get an exact answer.

This is also the first chapter where a graph genuinely earns its place in your working, rather than being optional decoration. A student who has only ever seen the equation $x^2 + y^2 - 6x + 4y - 12 = 0$ has to do real work to picture a circle; a student who has seen it plotted immediately has a mental anchor for what “centre” and “radius” mean before the algebra even starts.

How to use these notes

Each section keeps the same three-part structure as Chapters 1 and 2 — **Concept**, **Procedure**, **Worked example** — and most sections now also include a **graph** showing the shape being discussed, generated directly from the worked example's numbers. Sketch your own rough version of each graph as you work through a question, even when a full accurate plot isn't required for marks: it catches sign errors and unreasonable answers before they're written down.

3.1 Straight lines

Concept

A straight line is completely determined by its gradient (how steep it is, and in which direction) and one point it passes through. The gradient between two points (x_1, y_1) and (x_2, y_2) is $(y_2 - y_1)/(x_2 - x_1)$ — a ratio of “how far up” to “how far across”, which is why it's unaffected by which of the two points you call “first”.

This is the gradient of a straight line joining two fixed points. In Chapter 7 (Differentiation) you will meet the gradient of a curve at a single point, found as the limit of this same ratio as the two points move together — it is worth remembering that link when you get there, since the idea of gradient itself does not change, only how it is calculated.

Two facts about gradients drive almost every question in this section:

- **Parallel lines have equal gradients** — they point in exactly the same direction, so they never meet.
- **Perpendicular lines have gradients that multiply to -1** (equivalently, one gradient is the negative reciprocal of the other) — rotating a direction by 90° flips and inverts its steepness.

Procedure

1. Find the gradient using the two given points (or read it directly from an equation in the form $y = mx + c$).

- If you need a parallel line, use the same gradient; if you need a perpendicular line, use the negative reciprocal.
- Use the point-gradient form, $y - y_1 = m(x - x_1)$, with the gradient and any known point on the required line.
- Rearrange into the form requested (often $ax + by + c = 0$ with integer coefficients).

Worked example

EXAMPLE 3.1 (6 MARKS)

A is the point (1, 2) and B is the point (5, -4). Find the equation of the perpendicular bisector of AB, giving your answer in the form $ax + by + c = 0$, where a, b and c are integers.

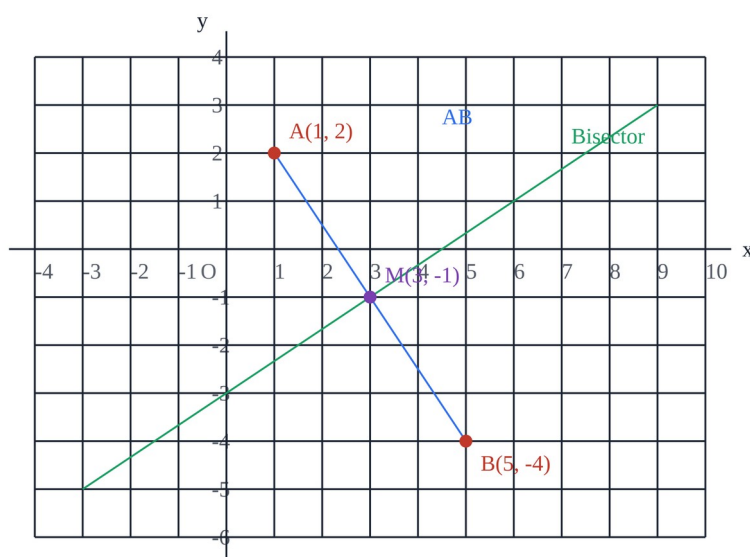
Step 1 Find the midpoint of AB, since the perpendicular bisector must pass through it: midpoint = $((1+5)/2, (2-4)/2) = (3, -1)$. (B1)

Step 2 Find the gradient of AB: $(-4 - 2)/(5 - 1) = -6/4 = -3/2$. (M1 A1)

Step 3 The perpendicular gradient is the negative reciprocal: $2/3$. (M1)

Step 4 Use the point-gradient form with the midpoint: $y - (-1) = (2/3)(x - 3)$. (M1)

Answer Rearrange: $3y + 3 = 2x - 6$, so $2x - 3y - 9 = 0$. (A1)



AB and its perpendicular bisector

The graph shows line segment AB and the perpendicular bisector crossing it at right angles through the midpoint M — notice they meet exactly once, and exactly where the algebra says they should.

EXAM TIP

Students often find the gradient of AB correctly but then forget to take the *negative reciprocal* for the perpendicular line, instead reusing the same gradient (which would make it parallel, not perpendicular) or just flipping the fraction without changing the sign. Drilling the phrase “flip and negate” alongside the picture of a 90° rotation helps it stick.

3.2 Circles

Concept

A circle is the set of all points a fixed distance (the radius) from a fixed point (the centre). Applying the distance formula to a general point (x, y) and a centre (a, b) gives $(x - a)^2 + (y - b)^2 = r^2$ directly — this equation is not a rule to memorise, it's simply “distance from (x, y) to the centre equals r ”, squared to remove the awkward square root.

When a circle's equation is expanded, it appears in the form $x^2 + y^2 + 2fx + 2gy + c = 0$ rather than the neat centre-radius form. Recovering the centre and radius from this expanded form is exactly the completing-the-square technique from Chapter 2, applied twice — once for the x terms, once for the y terms.

Procedure

1. If the equation is expanded, group the x terms and y terms, then complete the square on each group separately.
2. Rearrange into the form $(x - a)^2 + (y - b)^2 = r^2$.
3. Read off the centre (a, b) directly, and take the square root of the right-hand side to find r .

Worked example

EXAMPLE 3.2 (6 MARKS)

A circle has equation $x^2 + y^2 - 6x + 4y - 12 = 0$. (a) Find its centre and radius. (b) Show that the point $P(7, 1)$ lies on the circle.

(a) Step 1 Group the x and y terms: $(x^2 - 6x) + (y^2 + 4y) - 12 = 0$.

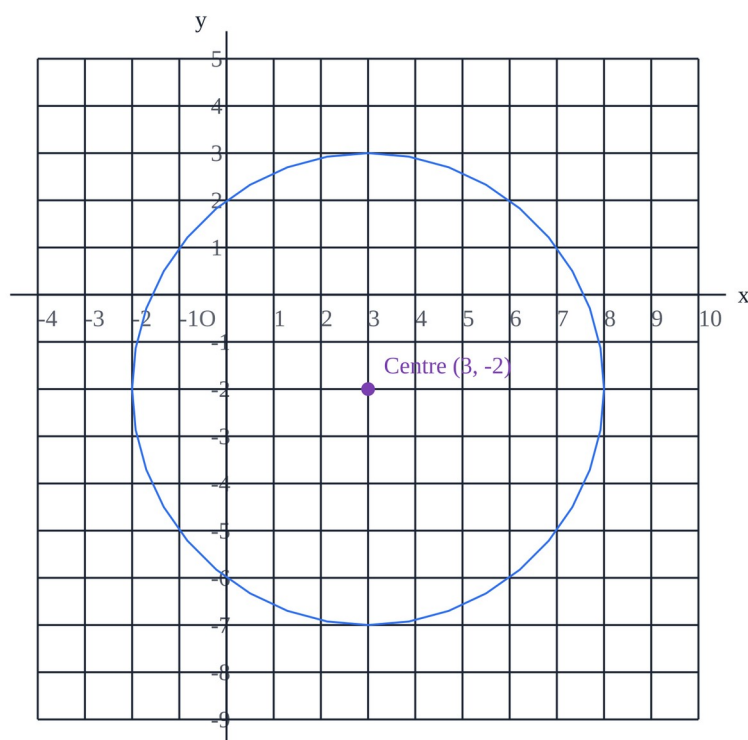
(a) Step 2 Complete the square on each group: $(x - 3)^2 - 9 + (y + 2)^2 - 4 - 12 = 0$. (M1 A1)

(a) Step 3 Rearrange: $(x - 3)^2 + (y + 2)^2 = 25$. (A1)

(a) Step 4 The centre is $(3, -2)$ and the radius is $\sqrt{25} = 5$. (A1)

(b) Step 1 Substitute $P(7, 1)$ into the centre-radius form: $(7 - 3)^2 + (1 - (-2))^2 = 4^2 + 3^2 = 16 + 9 = 25$. (M1)

Answer Since this equals $r^2 = 25$, P lies on the circle. (A1)



Circle plotted from its equation

EXAM TIP

The most common error here is a sign slip when completing the square on the y terms: $(y + 2)^2$ expands to $y^2 + 4y + 4$, so you must subtract 4 (not add it) to keep the equation balanced. Expand your completed-square form back out as a check — it should reproduce the original equation exactly.

3.3 Circle geometry: tangents and circle theorems in coordinates

Concept

This section applies two circle facts you likely already know from GCSE circle theorems, now proved and used with coordinates and gradients instead of angle-chasing on a diagram:

- **A tangent to a circle is perpendicular to the radius drawn to the point of contact.** This makes sense physically: the tangent is the direction you'd be moving in if you were travelling around the circle at that exact instant, and that direction of travel is always at right angles to the direction pointing to the centre.
- **The angle in a semicircle is a right angle.** If AB is a diameter and P is any other point on the circle, angle $APB = 90^\circ$. In coordinates, this becomes a gradient statement: the gradients of PA and PB multiply to -1 .

Procedure: finding a tangent

1. Find the gradient of the radius from the centre to the given point on the circle.

- The tangent's gradient is the negative reciprocal of this (since tangent $\perp$ radius).
- Use the point-gradient form with the given point and this gradient to find the tangent's equation.

Procedure: using the semicircle theorem

- Identify the diameter's endpoints and the point being tested.
- Find the gradient from each diameter endpoint to the test point.
- Multiply the two gradients: if the result is -1 , the angle at the test point is 90° , confirming (by the converse of the theorem) that the point lies on the circle with that diameter.

Worked example

EXAMPLE 3.3 (6 MARKS)

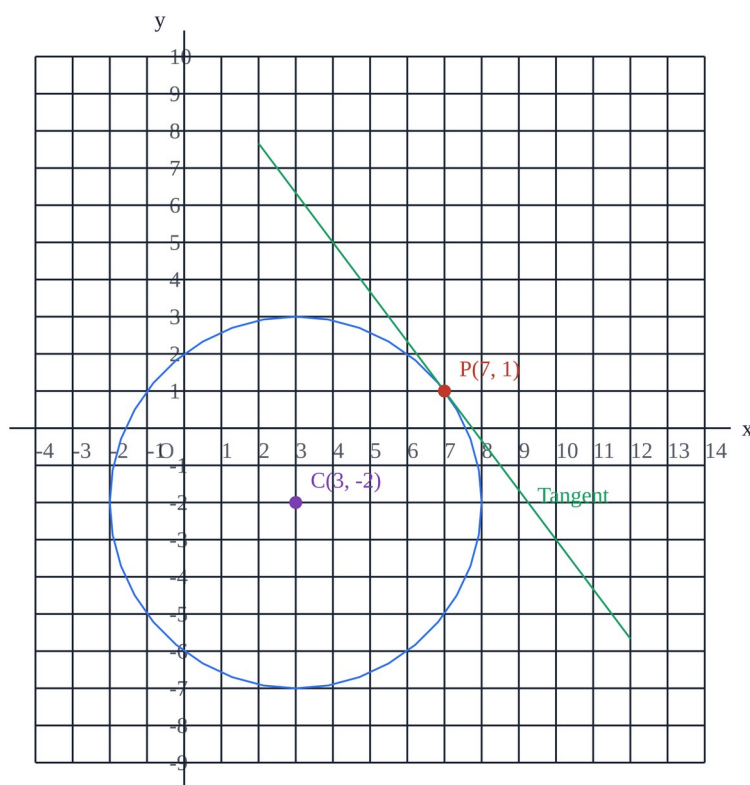
Using the circle from section 3.2 (centre $(3, -2)$, radius 5), find the equation of the tangent to the circle at the point $P(7, 1)$.

Step 1 Find the gradient of the radius CP , from the centre $(3, -2)$ to $P(7, 1)$: $(1 - (-2))/(7 - 3) = 3/4$. (M1 A1)

Step 2 The tangent gradient is the negative reciprocal: $-4/3$. (M1 A1)

Step 3 Use the point-gradient form: $y - 1 = -(4/3)(x - 7)$. (M1)

Answer Multiply through by 3: $3(y - 1) = -4(x - 7)$, so $3y - 3 = -4x + 28$, so $4x + 3y - 31 = 0$. (A1)



Circle and its tangent at P

The graph shows the tangent touching the circle at exactly one point, P, at a right angle to the radius CP — this is what makes it a tangent rather than a chord.

EXAM TIP

For the semicircle theorem, remember the result only follows automatically for a diameter — if AB is any chord (not a diameter), the angle at a point on the major arc is not generally 90° . Checking “is AB actually a diameter?” before applying the theorem is a common thing to skip under exam pressure.

3.4 Parametric equations: converting to Cartesian form

Concept

So far in this chapter, every curve has been described by a single equation directly linking x and y . A **parametric** description instead gives x and y each as separate functions of a third variable, the **parameter** (usually t or θ) — as the parameter sweeps through its values, the point $(x(t), y(t))$ traces out the curve, like following the position of a moving object over time.

Converting from parametric to Cartesian form means **eliminating the parameter**: finding a way to combine the two equations so that t (or θ) disappears, leaving a direct relationship between x and y only. There's no single method — the right approach depends on how the parameter appears, which is why this section shows two different techniques.

Procedure: eliminating the parameter

Method 1 — direct substitution (used when one equation can be rearranged for t):

1. Rearrange one equation to make t the subject.
2. Substitute this expression for t into the other equation.

Method 2 — using a trig identity (used when x and y involve $\sin$ and $\cos$ of the same angle):

1. Rearrange to isolate $\cos \theta$ (or $\sin \theta$) in one equation and $\sin \theta$ (or $\cos \theta$) in the other.
2. Substitute both into the identity $\sin^2 \theta + \cos^2 \theta = 1$.

Worked example

EXAMPLE 3.4 (6 MARKS)

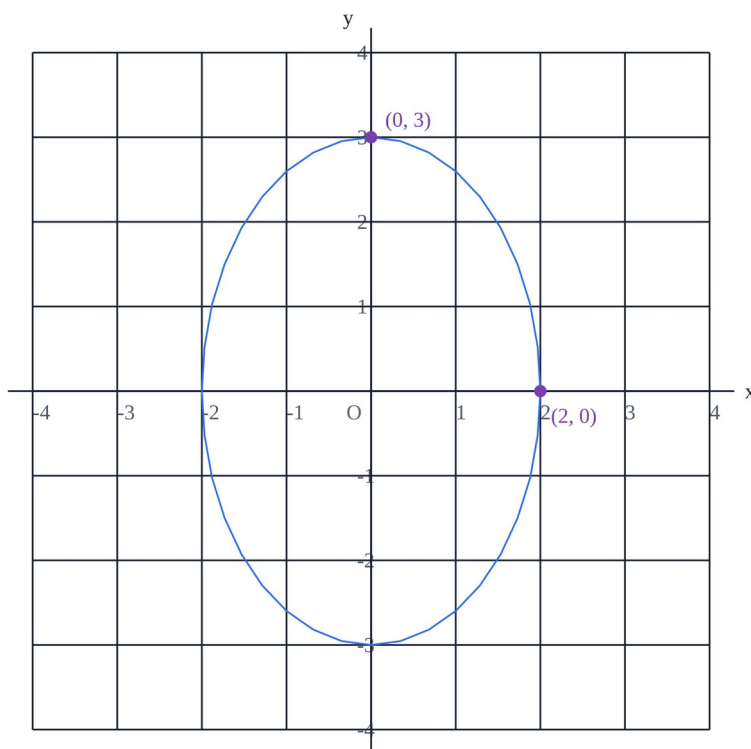
A curve has parametric equations $x = 2\cos\theta$ and $y = 3\sin\theta$, for $0 \leq \theta < 360^\circ$. Find the Cartesian equation of the curve, and describe the shape it represents.

Step 1 Rearrange each equation to isolate the trig function: $\cos\theta = x/2$ and $\sin\theta = y/3$. (M1)

Step 2 Substitute into the identity $\sin^2\theta + \cos^2\theta = 1$: $(y/3)^2 + (x/2)^2 = 1$. (M1 A1)

Step 3 Rewrite neatly: $x^2/4 + y^2/9 = 1$. (A1)

Answer This is the equation of an ellipse centred at the origin, stretching 2 units left/right and 3 units up/down from the centre. (B1 B1 — correct shape and correct extent identified)



Parametric curve plotted from $x=2\cos\theta$, $y=3\sin\theta$

The graph confirms the shape: notice it is 3 units tall but only 2 units wide, matching the 9 (from $y^2/9$) and 4 (from $x^2/4$) in the Cartesian equation — the larger denominator always corresponds to the direction the ellipse stretches further.

EXAM TIP

Remember this identity-based method is only available when the parameter appears as an angle inside sin and cos. If instead they see something like $x = t^2$ and $y = 2t$, method 1 (rearranging $y = 2t$ to get $t = y/2$, then substituting into $x = t^2$) is the one to reach for — recognising which method applies is the real skill being tested, not the substitution itself.

3.5 Parametric equations: curve sketching and points on a curve

Concept

Because a parametric curve is generated by a moving parameter, it can trace shapes that an ordinary $y = f(x)$ equation cannot easily describe — including curves that cross over themselves, loop back, or aren't functions at all (where a single x -value corresponds to more than one y -value). Sketching a parametric curve directly, without converting to Cartesian form first, means building up a table of (x, y) points for a range of parameter values and plotting them in order — the order matters, because it shows the direction the curve is traced in as the parameter increases.

Finding where a parametric curve crosses a given line or the axes, or finding the parameter value at a specific point, is really just solving simultaneous equations where the parametric equations themselves are two of the equations involved.

Procedure: sketching from a table of values

1. Choose a sensible range and spacing of parameter values (the question or context usually restricts the range).
2. Calculate x and y for each chosen value, keeping the results in a table.
3. Plot the points in the order the parameter increases, and join them smoothly — the direction of travel (shown with an arrow) is often worth marking.

Procedure: finding where a curve meets an axis

To find where a curve meets the x -axis, set $y(t) = 0$ and solve for t , then substitute back into $x(t)$. To find where it meets the y -axis, do the reverse.

Worked example**EXAMPLE 3.5 (7 MARKS)**

A curve has parametric equations $x = t^2 - 1$ and $y = t^3 - t$, for $-2 \leq t \leq 2$. (a) Find the coordinates of the points where the curve crosses the x -axis. (b) Show that the curve passes through the origin twice, for two different values of t .

(a) Step 1 The curve meets the x -axis when $y = 0$: $t^3 - t = 0$. (M1)

(a) Step 2 Factorise: $t(t^2 - 1) = t(t - 1)(t + 1) = 0$, so $t = 0, 1$ or -1 . (M1 A1)

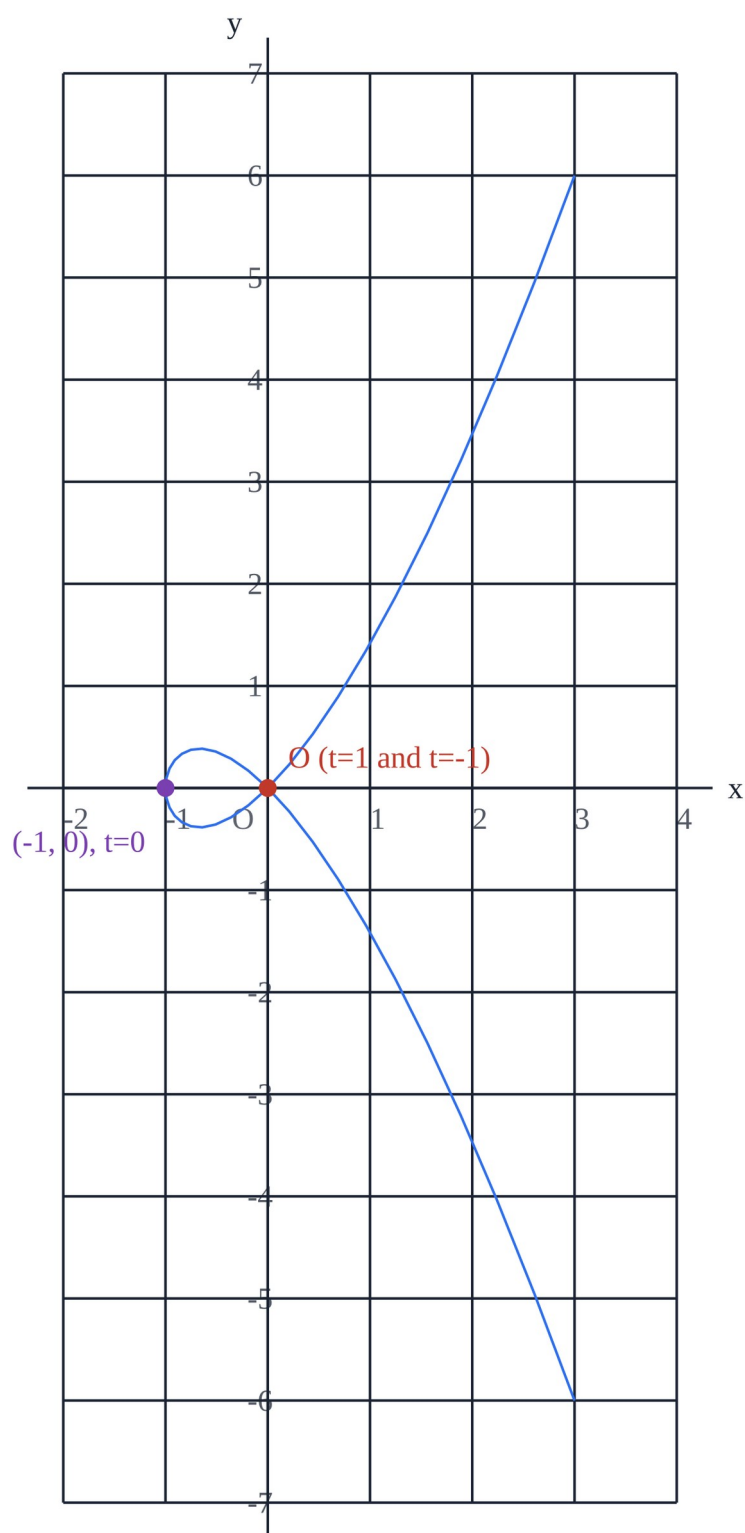
(a) Step 3 Find the corresponding x -values: at $t = 0$, $x = -1$; at $t = 1$, $x = 0$; at $t = -1$, $x = 0$.

(a) Step 4 The curve crosses the x -axis at $(-1, 0)$ and $(0, 0)$, with $(0, 0)$ reached at both $t = 1$ and $t = -1$. (A1)

(b) Step 1 The curve passes through the origin when $x = 0$ AND $y = 0$ simultaneously. From $x = t^2 - 1 = 0$, $t = 1$ or $t = -1$. (M1)

(b) Step 2 Check y at both: at $t = 1$, $y = 1 - 1 = 0$ ✓; at $t = -1$, $y = -1 - (-1) = 0$ ✓. (M1)

Answer Both values of t give the point $(0, 0)$, so the curve genuinely passes through the origin twice — once on its way in, once on its way out. (A1)



Parametric curve $x=t^2-1$, $y=t^3-t$

The graph shows exactly this: the curve crosses itself at the origin, forming a loop for $-1 \leq t \leq 1$, then continues outward for $|t| > 1$. This self-crossing is something an equation of the form $y = f(x)$ could never produce, since that would require $x = 0$ to give two different y -values at once.

EXAM TIP

Part (b) is the clearest illustration of why parametric curves are genuinely different from Cartesian ones: ask yourself what it would mean for an ordinary function $y = f(x)$ to pass through the same point twice, and you should immediately see it's impossible — a function gives exactly one output per input. The parameter is what allows the curve to revisit a point, by passing through it at two different “moments”.

Chapter summary

Section	Core skill	Key fact to anchor it
3.1 Straight lines	Gradient, point-gradient form	Parallel: equal gradients. Perpendicular: gradients multiply to -1
3.2 Circles	Completing the square (twice)	$(x-a)^2 + (y-b)^2 = r^2$ is just “distance from centre = r ”, squared
3.3 Circle geometry	Tangent $\perp$ radius; angle in a semicircle	Tangent gradient $\times$ radius gradient = -1
3.4 Parametric $\rightarrow$ Cartesian	Eliminating the parameter	Substitution, or the $\sin^2 + \cos^2 = 1$ identity
3.5 Parametric sketching	Plotting a table of (x,y) points in order	A parametric curve can cross itself; $y=f(x)$ never can

Common exam mistakes across this chapter

- **Forgetting to negate when finding a perpendicular gradient** — flipping the fraction is not enough on its own; the sign must change too.
- **Sign errors completing the square**, especially on the y terms when the coefficient is already negative.
- **Confusing a tangent problem with a general line problem** — always find the radius gradient first, since the tangent's gradient depends on it.
- **Applying the semicircle theorem to a chord that isn't a diameter.**
- **Trying to substitute one parametric equation directly into the Cartesian answer** rather than actually eliminating the parameter algebraically.
- **Sketching a parametric curve as if it must be a function** — forgetting that the same x -value can legitimately appear twice with different y -values (or vice versa).

Suggested practice

Because this chapter rewards picture-based reasoning as much as algebra, sketch a rough graph before you start the algebra on a past-paper question, purely as a sense check, then compare it to your final algebraic answer at the end. A mismatch between the rough sketch and the algebraic answer is one of the fastest ways to catch an error before submitting.

You should now be able to

- Find the equation of a line given points/gradients, including parallel and perpendicular lines
- Find a circle's centre and radius by completing the square, and work with tangents and circle theorems in coordinates
- Convert a parametric curve to Cartesian form, and sketch or analyse a parametric curve directly